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3. Transduction

Zinder and Lederberg (1952) discovered Transduction in *Salmonella typhimurum*. Phage mediated DNA transfer is called Transduction (Figure 1.15).

Transduction is of two types

(i) Generalized Transduction (ii) Specialized or Restricted Transduction

(i) Generalized Transduction

The ability of a bacteriophage to carry genetic material of any region of bacterial DNA is called Generalised transduction.

(ii) **Specialized or Restricted Transduction**

The ability of the bacteriophage to carry only a specific region of the bacterial DNA is called specialized or restricted transduction.

1.4.7 Economic importance of Bacteria

Bacteria are both beneficial and Harmful. The beneficial activities of bacteria are given in Table 1.7.

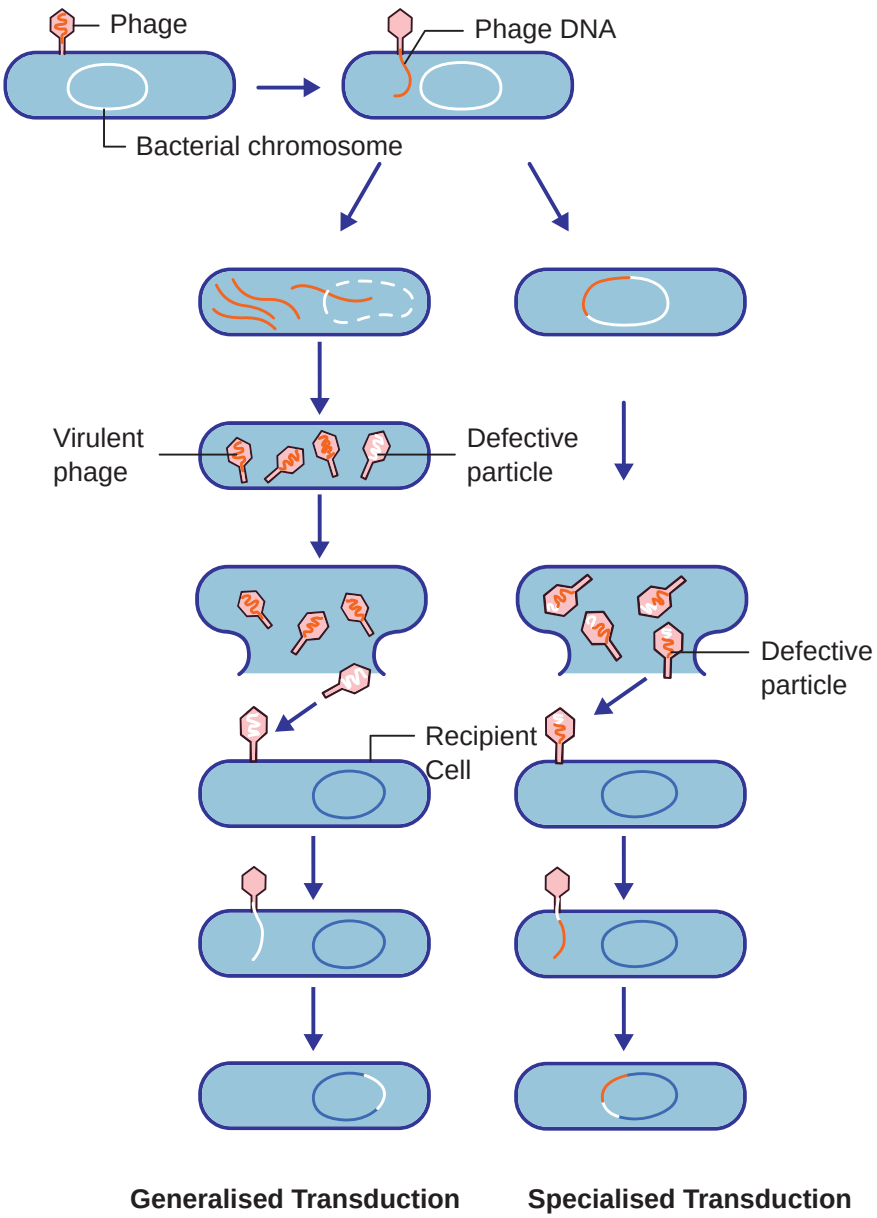


Figure 1.15: Transduction in Bacteria